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# Adebanji Adelowo

**Applied Mathematician · Machine Learning Researcher**

Selected research and applied engineering projects spanning phase-field PDE modelling, scientific machine learning, inverse and neural rendering, medical imaging, and signal processing for aviation systems. Each entry lists the problem addressed, the technical approach, and the result, with a link to the working code.

## Research Projects

### Phase-Field Modelling of Multiphase-Flow PDEs

**SCIENTIFIC COMPUTING · PDES · PHD TRACK**

*A PDE-based framework for modelling vapour-liquid phase change and coupled flow/heat-transfer dynamics, built for 3D extensibility from the outset.*

**PROBLEM**

Modelling the coupled interface, flow, and heat-transfer physics of boiling (vapour-liquid phase change under turbulent flow) with a PDE formulation that stays numerically tractable and portable to 3D.

**APPROACH**

A conservative Allen-Cahn phase-field model is coupled with incompressible Navier-Stokes and an energy equation containing a latent-heat source term. The resulting pressure Poisson equation has constant coefficients and is solved directly with FFTs, a property that holds in 3D and is the main motivation for the method.

**VALIDATION**

The 2D/1D prototype is benchmarked against two analytical solutions: bubble growth under a prescribed vaporisation rate, and the 1D Stefan problem. The bubble-growth benchmark reproduces the predicted linear growth rate; the Stefan-problem benchmark matches at early times, with growing deviation at late times attributable to the current simplified vaporisation-rate model. A 3D solver has been implemented; 3D validation and a probe-based vaporisation-rate model are ongoing.

**github.com/AdebanjiAdelowo/boiling-phasefield-3d**

### Optimal Mixing of Passive Scalars

**APPLIED MATHEMATICS · PDES · M.SC. THESIS**

*A Python implementation of the Lin-Thiffeault-Doering optimal-mixing scheme for passive scalars, deriving the optimal velocity from first principles and validating it by pseudo-spectral simulation.*

**PROBLEM**

How efficiently can an incompressible, enstrophy-constrained velocity field mix a passive scalar, and what are the theoretical limits on that mixing rate?

**APPROACH**

The Lin-Thiffeault-Doering optimal mixing velocity is derived from first principles via a constrained-optimisation argument, then simulated on the 2-torus with a pseudo-spectral solver, Leray projection, and RK45 time integration.

**RESULT**

Simulation confirms exponential $H^{-1}$ decay across initial conditions and reproduces published numerical results to four decimal places, including a resolution-sensitivity study (N=32 vs. N=64) showing the fitted mixing-rate exponent is itself resolution-dependent.

**Thesis (PDF)** · **github.com/AdebanjiAdelowo/Master-Thesis**

### Physics-Informed Neural Network: 1D Advection-Diffusion

**SCIENTIFIC MACHINE LEARNING**

*A PINN that learns the solution of a linear PDE directly from its residual, without any labelled interior data.*

**PROBLEM**

Approximating the solution of a linear PDE (1D advection-diffusion) with a neural network trained without labelled interior data.

**APPROACH**

A PyTorch PINN enforces the PDE residual, initial condition, and periodic boundary condition simultaneously via automatic differentiation, trained on collocation points with Adam.

The trained network matches the known analytical solution across the full space-time domain.

**~$5\times10^{-3}$** RELATIVE $L^2$ ERROR

**github.com/AdebanjiAdelowo/pinn-advection-diffusion**

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### 3D Liver Segmentation: Abdominal CT

**MEDICAL IMAGING**

*A 3D U-Net trained end-to-end for volumetric liver segmentation from abdominal CT.*

**PROBLEM**

Automatic segmentation of liver structures from abdominal CT volumes for downstream clinical and computational analysis.

**APPROACH**

A 3D U-Net trained end-to-end on 131 MSD Task03 liver CT cases, with foreground-biased patch sampling, a combined soft-Dice and BCE loss, cosine learning-rate annealing, and Gaussian sliding-window inference.

Best validation performance after 200 epochs on a single Kaggle T4 GPU.

**0.9886** VALIDATION DICE **≤1 mm** HD95

**github.com/AdebanjiAdelowo/abdominal-ct-segmentation**

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### Differentiable PBR: Inverse Material Optimisation

**INVERSE RENDERING**

*Recovering physical material properties from images by backpropagating through the rendering equation itself.*

**PROBLEM**

Recovering spatially-varying surface material properties (albedo, normal, roughness, metallic) from only a set of multi-view photographs.

**APPROACH**

A pure-PyTorch differentiable renderer implements the Cook-Torrance BRDF (GGX normal distribution, Schlick Fresnel, Smith geometry term), optimising material maps by gradient descent against 8 reference views, with no external differentiable-rasterisation library.

Material maps converge after 2,000 optimisation steps.

**~$1\times10^{-6}$** MSE VS TARGET RENDERS

**github.com/AdebanjiAdelowo/diff-pbr**

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### NeRF vs TensoRF: Neural Radiance Fields

**NEURAL RENDERING**

*Comparing an implicit MLP scene representation against an explicit tensor-decomposed one, implemented from scratch.*

**PROBLEM**

Reconstructing a continuous 3D scene representation from posed 2D images, and understanding the trade-off between an implicit MLP representation and an explicit tensor-decomposed one.

**APPROACH**

Vanilla NeRF and TensoRF (vector-matrix decomposition, Chen et al. 2022) are implemented from scratch in PyTorch, including an MPS-safe gather-based interpolation scheme for Apple Silicon, and compared under matched conditions.

TensoRF reaches a higher PSNR than vanilla NeRF in half the training iterations (15k vs 30k).

**17.2 dB** PSNR (TENSORF) **+3 dB** VS VANILLA NERF

github.com/AdebanjiAdelowo/nerf-tensorf

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## Applied ML & Engineering Projects

### Aircraft Engine Monitoring via Audio DSP

**SIGNAL PROCESSING · EMBEDDED SYSTEMS**

*Real-time engine RPM estimation from microphone audio alone, with no tachometer hardware.*

**PROBLEM**

Estimating aircraft engine RPM and operational status in real time without dedicated tachometer hardware.

**SYSTEM**

FFT-based signal processing runs on microphone audio on a Raspberry Pi, which transmits extracted RPM and status over UART to an ESP32 controller for real-time display.

**Validation:** tested against a real Tecnam P92 / Rotax 912 engine, demonstrating accurate RPM extraction from acoustic signal alone.

github.com/AdebanjiAdelowo/aircraft-engine-monitor

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### INTUOS Flight Data Monitoring Dashboard

**AVIATION · FULL-STACK ML DEPLOYMENT**

*A production monitoring dashboard serving ML-based flight-phase classifiers over live telemetry.*

**PROBLEM**

Giving operations teams real-time visibility into flight telemetry, engine performance, and phase of flight for safety analytics.

**SYSTEM**

A full-stack platform (FastAPI and IBM DB2 backend, React/Vite frontend, Dockerised deployment) serving ML-based flight-phase classifiers over live telemetry data.

**Deployment:** running as the operational monitoring dashboard used internally at Intuos Srl for flight performance analysis.

github.com/AdebanjiAdelowo/intuos-fdm-dashboard

### LLM Engineering Portfolio: End-to-End AI Systems

**GENERATIVE AI · NLP**

*Nine end-to-end LLM and generative AI systems, built as production-oriented pipelines rather than prototype API calls.*

**PROBLEM**

Building production-oriented LLM and generative AI systems (retrieval-augmented generation, structured data access, model compression) rather than prototype API calls.

**SYSTEM**

Nine end-to-end projects, including a Medical RAG Assistant (LangChain, ChromaDB, conversational memory), a two-stage Enterprise NL-to-SQL pipeline, LoRA/QLoRA parameter-efficient fine-tuning, and a compression pipeline (Taylor-gradient pruning plus knowledge distillation) applied to a FinBERT financial-sentiment classifier.

The distilled FinBERT model exceeds the 110M-parameter baseline's 84.5% accuracy and 0.843 F1.

**96.9%** ACCURACY / F1 (DISTILLED) **88M** PARAMS (VS 110M BASELINE)

**github.com/AdebanjiAdelowo/ai-projects-portfolio**

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## Additional Technical Work (supporting breadth in signal processing, optimisation, and generative modelling)

- **Dipole Subsurface Scattering** (**sss-dipole**): differentiable dipole BSSRDF (Donner and Jensen 2005) in PyTorch, parameterised for all six Fitzpatrick skin types.
- **Image Inpainting** (**Image_inpainting**): Douglas-Rachford optimisation for variational, TV-regularised image restoration.
- **Image Denoising** (**Image_denoising_using_ResNet**): residual channel-attention blocks for blind greyscale denoising, approximately 26 to 28 dB PSNR on LFW.
- **nano-gpt** (**nano-gpt**): a GPT-style transformer language model built from scratch in PyTorch, trained character-level on Tiny Shakespeare.
- **Whole-Body Diffusion Generator** (**whole-body-diffusion-generator**): IP-Adapter FaceID and SDXL face-guided portrait generation, served on serverless GPU.
- **Roof Segmentation** (**Roof-segmentation**): U-Net based aerial image segmentation for rooftop detection from satellite imagery.

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All results above are as reported in each project's own repository (code, README, and generated figures); no metrics have been added or extrapolated beyond what each repository documents. Full portfolio with visual results: adebanjioluwatimileyin.github.io/projects.html